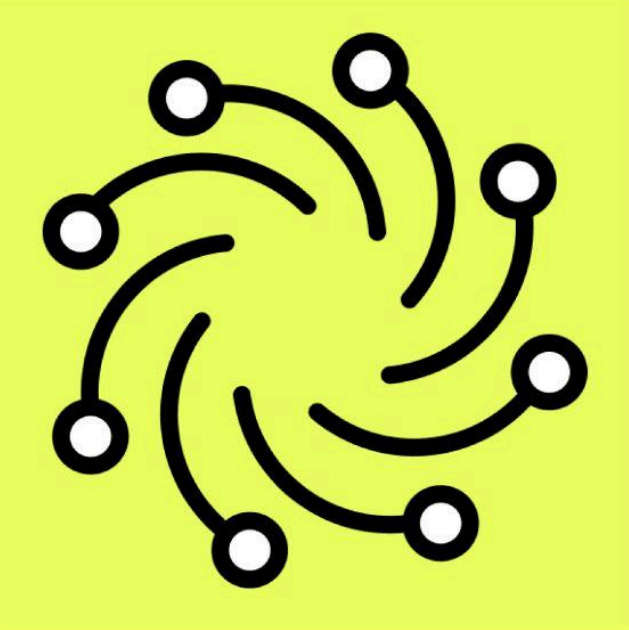
audita



MageLabs DEX

Manual and Automated Testing Report

June 11th, 2025

Prepared for:
Magelabs - Decentralized Exchange on Solana (*magelabs.xyz*)

Presented By:
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Document

The contents of this document may include confidential information pertaining to the IT systems, intellectual property, and possible vulnerabilities that the Client may possess. The report that contains this confidential information can be utilized internally by the Client, and shown to Investor parties.

Network: Solana, Devnet
Programming language: Rust
Method: Manual and Automated Testing
Client Website: https://magelabs.xyz
Timeline: 04 June 2025 - 11 June 2025

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Summary

★ The magelabs-client CLI tool demonstrated correct behavior in serializing and transmitting user inputs to the on-chain program.

★ The core lifecycle of liquidity provision, including depositing assets, receiving LP tokens, and withdrawing the full balance of those LP tokens, was successfully validated.

★ Administrative functions, such as updating a pool's operational status to enable or disable swaps, also performed as expected.

★ The test of the fee accrual mechanism confirmed that trading fees generated within a pool are correctly attributed to liquidity providers' positions and realized upon withdrawal.

Findings Requiring Further Review

- ★ Swaps below a certain size consistently failed due to the interaction between the input amount and the pool's fixed-fee structure, resulting in a ***RequireGtViolated*** error. **This indicates the current fee model may not be suitable for all trade sizes.**

- ★ Additionally, when a swap was attempted on a pool where trading was disabled via the status flag, the program returned a ***PoolNotOpenYet*** error instead of a more specific status-related error, which could **need further clarification to improve diagnostics for developers and users.**

If the findings listed above are considered problematic by Client, our team can commit a fix as per Client's wishes.

Automated Testing

Automated Tests - Github

Find all passing automated tests on Github:

https://github.com/qu0laz/magelabs-gtm/tree/Audita/programs/mage/tests

Documentation

We recommend this report, as well as specific information from this report to be included in MageLabs DEX's official protocol **Documentation**.

Manual Testing

Basic Swap (MAGE -> USDC)

tx:
https://solscan.io/tx/3cx4s88wkqi1Y3TnboSsNjoVRALqACegcqZkZCf11orLFyNHxZesQENVFfpuLPxkgD1jcu27ThEef1zogNNmeVpC?cluster=devnet

Command run: cargo run --bin magelabs-client -- amm-swap-base-in --pool-id ANqMZnntf9VnWTt5PA8FNLV193GwmXvgF9wPydZaEC1q --input-mint 28ji9sTFSNaWrgE31TFM7vWu5WJZabMoDXxnd9ZCAxxa --amount-in 10000000000 --minimum-amount-out 8000000

Keypair: admin.json (44Bk76bDAP5oKS14G6T2uHWoWpKsdL9hkLzvhoMXinkQ)

Expected result: The swap succeeds. The admin wallet's MAGE balance decreases by 10, and its USDC balance increases. The pool's MAGE vault increases by 10, and its USDC vault decreases.

Actual result: Success. MAGE wallet balance decreased by 10. USDC wallet balance increased by ~8.27. MAGE vault balance increased by 10. USDC vault balance decreased by ~8.27.

Deposit Liquidity

tx:
https://solscan.io/tx/2wayqvskpvhU2qyAvdDw7uMT7yPMmAe8Ng6QrPGpyKPywwZR9AQVp2G9TU54REqudNqb9QGimhW8t1b3TA3haJE?cluster=devnet

Command run: cargo run --bin magelabs-client -- amm-deposit --pool-id ANqMZnntf9VnWTt5PA8FNLV193GwmXvgF9wPydZaEC1q --lp-token-amount 50000000 --maximum-token-0-amount 5000000000 --maximum-token-1-amount 5000000

Keypair: admin.json (44Bk76bDAP5oKS14G6T2uHWoWpKsdL9hkLzvhoMXinkQ)

Expected result: The deposit succeeds. The admin wallet's MAGE and USDC balances decrease, and its LP token balance increases.

Actual result: Success. LP token balance increased by 50. MAGE and USDC balances decreased proportionally.

Withdraw Liquidity

tx:
https://solscan.io/tx/2YAVCFWMS4LP1V9rm37cHMiPExhNZyuynkbuFU9XW6EcWhygNR8vP93sZ3Y9q9SCtgFTj2ZBnniLFHcFZYRKkxjs?cluster=devnet

Command run: cargo run --bin magelabs-client -- amm-withdraw --pool-id ANqMZnntf9VnWTt5PA8FNLV193GwmXvgF9wPydZaEC1q --lp-token-amount 25000000 --minimum-token-0-amount 300000000 --minimum-token-1-amount 600000

Keypair: admin.json (44Bk76bDAP5oKS14G6T2uHWoWpKsdL9hkLzvhoMXinkQ)

Expected result: The withdrawal succeeds. The admin wallet's LP token balance decreases, and its MAGE and USDC balances increase.

Actual result: Success. LP token balance decreased by 25. MAGE and USDC balances increased proportionally.

Slippage Protection Setup (Large Swap to Shift Price)

tx:
https://solscan.io/tx/QmBZDexfyZGeDcQD6DQrt1GJwL9t7mWkHfJnAQYySj2uo3H7UEnDjmyPKvYkjMX6XNmk85NANMLzvd9e7YqNJXx?cluster=devnet

Command run: cargo run --bin magelabs-client -- amm-swap-base-in --pool-id ANqMZnntf9VnWTt5PA8FNLV193GwmXvgF9wPydZaEC1q --input-mint 28ji9sTFSNaWrgE31TFM7vWu5WJZabMoDXxnd9ZCAxxa --amount-in 1000000000000 --minimum-amount-out 500000

Keypair: admin.json (44Bk76bDAP5oKS14G6T2uHWoWpKsdL9hkLzvhoMXinkQ)

Expected result: The large swap succeeds, significantly changing the price ratio in the pool.

Actual result: Success. The transaction was confirmed.

Fee Accrual Setup (Admin Deposit)

tx:
https://solscan.io/tx/Y6td4LJpAKebUGC3LzRYJKXukRDqctbQkK4SYJAZ42DdZLpbRSkfKC6TYcdMnd8VYk5fctqsN1rKhbHNr8BZzB3?cluster=devnet

Command run: cargo run --bin magelabs-client -- amm-deposit --pool-id ANqMZnntf9VnWTt5PA8FNLV193GwmXvgF9wPydZaEC1q --lp-token-amount 100000000 --maximum-token-0-amount 1000000000000 --maximum-token-1-amount 1000000

Keypair: admin.json (44Bk76bDAP5oKS14G6T2uHWoWpKsdL9hkLzvhoMXinkQ)

Expected result: The admin wallet successfully deposits liquidity and receives 100 LP tokens.

Actual result: Success. The transaction was confirmed.

Fee Accrual Setup (Fund Secondary Wallet)

tx:
https://solscan.io/tx/4gJSANSHWREaMn61BigJ5UCiYtLGcgFV2UPpjZ32EAc98YPHKoYVJLi2VK33RZeHfjkb2h3bTLhvXZ4dWsMwWLE2?cluster=devnet

Command run: spl-token transfer AwveSwnX4fy7FZwdyDsoZ8Q83cTWQMAnGQpdgLxYrbib 100000000 42C3oz6vxfEPctPduEL5g5gJqTYa6UhSHtgukTJKvr1n

Keypair: admin.json (44Bk76bDAP5oKS14G6T2uHWoWpKsdL9hkLzvhoMXinkQ)

Expected result: The secondary wallet receives USDC to perform swaps.

Actual result: Success. The transaction was confirmed.

Fee Accrual Setup (Generate Fees via Swaps)

tx: 5 separate swap transactions were executed.

Command run: for i in {1..5}; do echo "=== Swap $i ===" && cargo run --bin magelabs-client -- amm-swap-base-in --pool-id ANqMZnntf9VnWTt5PA8FNLV193GwmXvgF9wPydZaEC1q --input-mint AwveSwnX4fy7FZwdyDsoZ8Q83cTWQMAnGQpdgLxYrbib --amount-in 10000000 --minimum-amount-out 1000000; done

Keypair: devnet-magelabs-keypair.json (42C3oz6vxfEPctPduEL5g5gJqTYa6UhSHtgukTJKvr1n)

Expected result: Multiple swaps are executed successfully, generating fees within the pool.

Actual result: Success. All 5 swap transactions were confirmed.

Fee Accrual Verification (Admin Withdraw)

tx: *https://solscan.io/tx/ptxc3UUDivFVkT8ZQoWd9QBYYwct6rkzfW3UAYrUCAJoojCxjGUjAVS3BEoLVi1jjphZT3b45viaPgEZJtbiGwD?cluster=devnet*

Command run: cargo run --bin magelabs-client -- amm-withdraw --pool-id ANqMZnntf9VnWTt5PA8FNLV193GwmXvgF9wPydZaEC1q --lp-token-amount 100000000 --minimum-token-0-amount 0 --minimum-token-1-amount 0

Keypair: admin.json (44Bk76bDAP5oKS14G6T2uHWoWpKsdL9hkLzvhoMXinkQ)

Expected result: The admin withdraws their 100 LP tokens and receives back their principal plus a share of the accrued swap fees.

Actual result: Success. The admin wallet received ~6.16 more MAGE and ~6.60 more USDC than their proportional share, confirming fee accrual.

Pool Creation

tx:
https://solscan.io/tx/5DFKj217ExNJzTJ1SaWpCssBQRyDjXxsxjigw4GEGYQ9YJPXaQLE9Ssb9W1PfsBmeawyEmP9c8kh8TdZpFnXLszV?cluster=devnet

Command run: cargo run --bin magelabs-client -- amm-initialize-pool --mint0 4Rpaynvauvrfh6kdDDm7cE1sjXvc7YJTd5pEyqedUtGB --mint1 DtdCacfTjpWKitbhbXNkV6QcbFzFYht1i6mk8GNsewQp --init-amount-0 100000000000 --init-amount-1 100000000 --open-time 1749612734 --fee-tier-index 1
Keypair: admin.json (44Bk76bDAP5oKS14G6T2uHWoWpKsdL9hkLzvhoMXinkQ)

Expected result: A new pool is created successfully with the specified parameters and fee tier.

Actual result: Success. The transaction was confirmed and a new pool ID (AKzsc2Y2zUXMXzwD4E222ptRMdpofG9BwmksTC1XiKJL) was created.

Pool Time-Lock Verification

tx:
https://solscan.io/tx/2yyyza2uZcXhNhSbUuvTtP4bq63dw62K3RcHv3Sa2Vm9q6Ticd4wxitDG4rAtWD2C4Aifuxs7P1usBuNPg4nt6T3?cluster=devnet

Command run: cargo run --bin magelabs-client -- amm-swap-base-in --pool-id AKzsc2Y2zUXMXzwD4E222ptRMdpofG9BwmksTC1XiKJL --input-mint 4Rpaynvauvrfh6kdDDm7cE1sjXvc7YJTd5pEyqedUtGB --amount-in 10000000000 --minimum-amount-out 9000

Keypair: admin.json (44Bk76bDAP5oKS14G6T2uHWoWpKsdL9hkLzvhoMXinkQ)

Expected result: After waiting for the open_time to pass, the swap should succeed.

Actual result: Success. The transaction was confirmed.

Pool Status Update (Disable Swaps)

tx:
https://solscan.io/tx/3BznnGHUV3YWo8zVz5mbUcZ84stLZoJpAHTPx7h2keje7NzjTqcJgtAqCvWANyRLPsDJamE8pHa8SZSiiv8T32m?cluster=devnet

Command run: cargo run --bin magelabs-client -- amm-update-pool-status --pool-id AKzsc2Y2zUXMXzwD4E222ptRMdpofG9BwmksTC1XiKJL --status 4

Keypair: admin.json (44Bk76bDAP5oKS14G6T2uHWoWpKsdL9hkLzvhoMXinkQ)

Expected result: The pool status is successfully updated to disable swaps.

Actual result: Success. The transaction was confirmed.

Pool Status Update (Re-enable Swaps)

tx:
https://solscan.io/tx/612bjvxHnixZVNN1VdVh36SFx6DEPjkbLPtRmk4aKcdedD7GjFH1gyMQMULphyaB9s64nNFn2iwxUfZ8SZPdnbMh?cluster=devnet

Command run: cargo run --bin magelabs-client -- amm-update-pool-status --pool-id ANqMZnntf9VnWTt5PA8FNLV193GwmXvgF9wPydZaEC1q --status 0

Keypair: admin.json (44Bk76bDAP5oKS14G6T2uHWoWpKsdL9hkLzvhoMXinkQ)

Expected result: The pool status is successfully updated to re-enable all operations.

Actual result: Success. The transaction was confirmed.

Full Liquidity Withdrawal

tx:
https://solscan.io/tx/5Zt1pMxSNkJDh9ihxrYxq3oxLk2S8z5H8P7SFwf3nLrkhYCQEaHQMmUFvo1WdUSEw7tJXENVBBWmdiNCm4Pun1vz?cluster=devnet

Command run: cargo run --bin magelabs-client -- amm-withdraw --pool-id ANqMZnntf9VnWTt5PA8FNLV193GwmXvgF9wPydZaEC1q --lp-token-amount 2661884905 --minimum-token-0-amount 0 --minimum-token-1-amount 0

Keypair: admin.json (44Bk76bDAP5oKS14G6T2uHWoWpKsdL9hkLzvhoMXinkQ)

Expected result: The user withdraws their entire LP token balance, resulting in a balance of 0.

Actual result: Success. The final check of the admin wallet's LP token account showed a balance of 0.

Swap on Re-enabled Pool

tx:
https://solscan.io/tx/4KHTqDDNxNGg5kztgzsqfr37fLwPKVt7VvPZFssb6YQVWsxkaPZskeTV7fnsqY8kLjia8uuX8z9mUckaZzeLb8hT?cluster=devnet

Command run: cargo run --bin magelabs-client -- amm-swap-base-in --pool-id ANqMZnntf9VnWTt5PA8FNLV193GwmXvgF9wPydZaEC1q --input-mint 28ji9sTFSNaWrgE31TFM7vWu5WJZabMoDXxnd9ZCAxxa --amount-in 10000000000 --minimum-amount-out 100

Keypair: admin.json (44Bk76bDAP5oKS14G6T2uHWoWpKsdL9hkLzvhoMXinkQ)

Expected result: The swap should succeed after the pool is re-enabled.

Actual result: Success. The transaction was confirmed.

Swap on New Pool (Large Amount)

tx:
https://solscan.io/tx/5YpwYeBEVMvPVswkTyeMw74DhpGfz118Rz9Nf7yruc2e21tqsvoXKWtRM8srrReai4GgFNWjX5XRkfH2zyuJkTRm?cluster=devnet

Command run: cargo run --bin magelabs-client -- amm-swap-base-in --pool-id AKzsc2Y2zUXMXzwD4E222ptRMdpofG9BwmksTC1XiKJL --input-mint 4Rpaynvauvrfh6kdDDm7cE1sjXvc7YJTd5pEyqedUtGB --amount-in 50000000000 --minimum-amount-out 1

Keypair: admin.json (44Bk76bDAP5oKS14G6T2uHWoWpKsdL9hkLzvhoMXinkQ)

Expected result: The swap should succeed with a large enough input to overcome the fixed fee.

Actual result: Success. The transaction was confirmed.

Disclaimer

This deliverable makes no statements or warranties on the security of the code. While we have conducted the testing and analysis to our best abilities and produced this report in line with latest industry developments, it is important to not rely on this report only. In order for contracts to be considered as safe as possible, the industry standard requires them to be checked by several independent testers and auditing bodies.

Smart contract platforms, their programming languages, and other software components are not immune to bugs and vulnerabilities that can cause issues, which is why this report cannot provide an absolute guarantee.